

# Introduction

Doom is a documentation development tool designed for internal use at Alauda, built on top of [rspress](#). It provides users with a rich set of built-in plugins for an out-of-the-box experience.

## TOC

Core Capabilities

Based on Markdown and its extension MDX

Get Started

## Core Capabilities

- Automatically generates a configurable weight (order) left sidebar
- Full-text static document search
- Multilingual support

## Based on Markdown and its extension MDX

MDX is a powerful content development approach. You can write Markdown files as you normally would while also using React components within the Markdown content:



```
// docs/index.mdx
import { CustomComponent } from './custom';

# Hello World

<CustomComponent />
```

For more details, you can refer to the [“Using MDX” documentation](#).

---

## Get Started

Let's [get started quickly](#) with Doom!

# Getting Started

## TOC

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## Creating a Project

First, you can create a new directory with the following command:

```
mkdir my-docs && cd my-docs
```

Run `npm init -y` to initialize a project. You can install doom using npm, yarn, or pnpm:

 npm    yarn    pnpm    bun

```
npm install -D @alauda/doom typescript
```

Then create files with the following commands:

```
# Create docs directories, default supports bilingual Chinese and English
mkdir docs/en && echo '# Hello World' > docs/en/index.md
mkdir docs/zh && echo '# 你好世界' > docs/zh/index.md
```

Add the following scripts to your `package.json` :

```
{
  "scripts": {
    "dev": "doom dev",
    "build": "doom build",
    "new": "doom new",
    "serve": "doom serve",
    "translate": "doom translate",
    "export": "doom export"
  }
}
```

Then initialize a configuration file `doom.config.yml` :

```
title: My Docs
```

Also create a `tsconfig.json` file with the following content:

```
{
  "compilerOptions": {
    "jsx": "react-jsx",
    "module": "NodeNext",
    "moduleResolution": "NodeNext",
    "noUnusedLocals": true,
    "noUnusedParameters": true,
    "resolveJsonModule": true,
    "skipLibCheck": true,
    "strict": true,
    "target": "ESNext",
  },
  "mdx": {
    "checkMdx": true,
  },
}
```

Finally, create a `global.d.ts` file with the following content:

```
/// <reference types="@alauda/doom/runtime" />
```

This allows you to safely use the global components provided by doom in `.mdx` files with type safety.

---

## CLI Tool

```
doom -h
```

```
# output
```

```
Usage: doom [options] [command]
```

```
Doctor Doom making docs.
```

#### Options:

|                                                |                                                                                                                                                             |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>-V, --version</code>                     | output the version number                                                                                                                                   |
| <code>-c, --config &lt;config&gt;</code>       | Specify the path to the config file                                                                                                                         |
| <code>-v &lt;version&gt;</code>                | Specify the version of the documentation, c                                                                                                                 |
| <code>-b, --base &lt;base&gt;</code>           | Override the base of the documentation                                                                                                                      |
| <code>-p, --prefix &lt;prefix&gt;</code>       | Specify the prefix of the documentation bas                                                                                                                 |
| <code>-f, --force [boolean]</code>             | Force to <ol style="list-style-type: none"> <li>1. fetch latest reference remotes or scaffo</li> <li>2. translate ignore hash equality check and</li> </ol> |
| <code>-i, --ignore [boolean]</code>            | Ignore internal routes (default: false)                                                                                                                     |
| <code>-d, --download [boolean]</code>          | Display download pdf link on nav bar (defau                                                                                                                 |
| <code>-e, --export [boolean]</code>            | Run or build in exporting PDF mode, `apis/*                                                                                                                 |
| <code>-I, --include &lt;language...&gt;</code> | Include <b>only</b> the specific language(s),                                                                                                               |
| <code>-E, --exclude &lt;language...&gt;</code> | Include all languages except the specific l                                                                                                                 |
| <code>-o, --out-dir &lt;path&gt;</code>        | Override the `outDir` defined in the config                                                                                                                 |
| <code>-r, --redirect &lt;enum&gt;</code>       | Whether to redirect to the locale closest t                                                                                                                 |
| <code>-R, --edit-repo [boolean url]</code>     | Whether to enable or override the `editRepo                                                                                                                 |
| <code>-a, --algolia [boolean alauda]</code>    | Whether to enable or use the alauda (docs.a                                                                                                                 |
| <code>-S, --site-url</code>                    | Whether to enable the siteUrl for sitemap g                                                                                                                 |
| <code>-n, --no-open</code>                     | Do not open the browser after starting the                                                                                                                  |
| <code>-h, --help</code>                        | display help for command                                                                                                                                    |

#### Commands:

|                                             |                                            |
|---------------------------------------------|--------------------------------------------|
| <code>dev [options] [root]</code>           | Start the development server               |
| <code>build [root]</code>                   | Build the documentation                    |
| <code>preview serve [options] [root]</code> | Preview the built documentation            |
| <code>new [template]</code>                 | Generate scaffolding from templates        |
| <code>translate [options] [root]</code>     | Translate the documentation                |
| <code>export [options] [root]</code>        | Export the documentation as PDF, `apis/**` |
| <code>lint [root]</code>                    | Lint the documentation                     |
| <code>help [command]</code>                 | display help for command                   |

For more configuration, please refer to [Configuration](#)

## Starting the Development Server

Run `yarn dev` to start the development server, the browser will automatically open the documentation homepage.

```
doom dev -h
```

```
# output
```

```
Usage: doom dev [options] [root]
```

```
Start the development server
```

```
Arguments:
```

```
  root
```

```
Root directory of the documentation
```

```
Options:
```

```
-H, --host [host]
```

```
Dev server host name
```

```
-P, --port [port]
```

```
Dev server port number
```

```
-l, --lazy [boolean]
```

```
Whether to enable `lazyCompilation` which could i
```

```
--no-lazy
```

```
Do not enable `lazyCompilation`
```

```
-h, --help
```

```
display help for command
```

## Production Build

Run `yarn build` to build the production code. After building, static files will be generated in the `dist` directory.

## Local Preview

Run `yarn serve` to preview the built static files. Note that if you used `-b` or `-p` parameters during build, you need to use the same `-b` and `-p` parameters during preview.

## Using Scaffolding Templates

Run `yarn new` to generate projects, modules, or documentation from scaffolding templates.

## Translating Documentation

```
doom translate -h
```

```
# output
```

```
Usage: doom translate [options] [root]
```

Translate the documentation

Arguments:

root                                      Root directory of the documentation

Options:

-s, --source <language> Document source language, one of en, zh, ru (default)

-t, --target <language> Document target language, one of en, zh, ru (default)

-g, --glob <path...> Glob patterns of source dirs/files to translate

-C, --copy [boolean] Whether to copy relative assets to the target directory

-h, --help display help for command

- The `-g, --glob` parameter is required. You can specify the directories or paths of files to translate, supporting `glob` syntax. Note that the parameter value must be quoted to avoid unexpected behavior from the command line parser. Examples:
  1. `yarn translate -g abc xyz` will translate all documents under `<root>/<source>/abc` and `<root>/<source>/xyz` to `<root>/<target>/abc` and `<root>/<target>/xyz` respectively.
  2. `yarn translate -g '*'` will translate all document files under `<root>/<source>`.
- The `-C, --copy` parameter is optional. It controls whether to copy local asset files to the target directory when the target file does not exist. The default is `false`, which means changing the asset reference path to the source path. Examples:
  - When this parameter is enabled:
    1. When translating `/<source>/abc.jpg`, it will copy `<root>/public/<source>/abc.jpg` to `<root>/public/<target>/abc.jpg` and modify the reference path in the document to `/<target>/abc.jpg`.
    2. For `<root>/<source>/abc.mdx` documents, references like `./assets/xyz.jpg` will copy `<root>/<source>/assets/xyz.jpg` to `<root>/<target>/assets/xyz.jpg`, and the image reference path remains unchanged.



3. For `<root>/<source>/abc.mdx` documents, references like `./assets/<source>/xyz.jpg` will copy `<root>/<source>/assets/<source>/xyz.jpg` to `<root>/<target>/assets/<target>/xyz.jpg` and modify the reference path in the document to `./assets/<target>/xyz.jpg`.

- When this parameter is not enabled:

1. When translating `/<source>/abc.jpg`, if `<root>/public/<target>/abc.jpg` exists, the reference path in the document will be changed to `/<target>/abc.jpg`; otherwise, the image reference path remains unchanged.

2. For `<root>/<source>/abc.mdx` documents, references like `./assets/<source>/xyz.jpg`, if `<root>/<target>/assets/<target>/xyz.jpg` exists, the reference path will be changed to `./assets/<target>/xyz.jpg`; otherwise, it will be changed to `../<source>/assets/<target>/xyz.jpg`.

### WARNING

Specifically, if you use `-g '*'` for full translation, the file lists of `source` and `target` directories will be compared, and unmatched `target` files except for `internalRoutes` will be automatically deleted.

### TIP

The translation feature requires the local environment variable `AZURE_OPENAI_API_KEY` to be configured. Please contact your team leader to obtain it.

You can control translation behavior in the document metadata:

```
i18n:
  title:
    en: DevOps Connectors
  additionalPrompts: 'The term Connectors in this document is a proper noun a
  disableAutoTranslation: false
  title: DevOps Connectors
```

For more configuration, please refer to [Translation Configuration](#)

## Exporting PDF

### WARNING

Please run the `yarn build` command before performing the export operation.

```
doom export -h
```

```
# output
```

```
Usage: doom export [options] [root]
```

```
Export the documentation as PDF, `apis/**` and `*/apis/**` routes will be ignored
```

```
Arguments:
```

```
  root                Root directory of the documentation
```

```
Options:
```

```
-H, --host [host]  Serve host name
-P, --port [port]  Serve port number (default: "4173")
-h, --help          display help for command
```

Run `yarn export` to export the documentation as a PDF file. Note that if you used `-b` or `-p` parameters during build, you need to use the same `-b` and `-p` parameters during export.

The export feature depends on [playwright](#) <sup>↗</sup>. For CI pipelines, please use `build-harbor.alauda.cn/frontend/playwright-runner:doom` as the base image for dependency installation and documentation build. Locally, you can set the following environment variable to speed up downloads:

```
.env.yarn
```

```
PLAYWRIGHT_DOWNLOAD_HOST="https://cdn.npmmirror.com/binaries/playwright"
```

Besides exporting a full-site unified PDF document, `doom` also supports exporting a single PDF file by specifying an entry point. For more configuration, please refer to [Document Export](#)

## Configuration

### Documentation Linting

```
doom lint -h
```

```
# output
```

```
Usage: doom lint [options] [root]
```

```
Lint the documentation
```

```
Arguments:
```

```
  root                Root directory of the documentation
```

```
Options:
```

```
-g, --glob <path...>  Glob patterns of source dirs/files to lint (default:
--no-cspell            Disable cspell linting
-h, --help            display help for command
```

`doom lint` is based on [ESLint](#) and [cspell](#). For a better experience in editors, you can install the corresponding plugins [ESLint](#) / [CSpell](#), then create the corresponding configuration files:

- `eslint.config.mjs` :

```
import doom from '@alauda/doom/eslint'

export default doom(new URL('docs', import.meta.url))
```

- `cspell.config.mjs` :

```
export { default } from '@alauda/doom/cspell'
```

We also agree that the `.cspell` folder in the current working directory ( `CWD` ) is used to store CSpell dictionary files. For example, you can create `.cspell/k8s.txt` :

k8s  
kubernetes

For more configuration, please refer to [Lint Configuration](#)

# Usage

# Configuration

---

## TOC

Configuration File

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---

## Configuration File

In most cases, we only need to use a static `yaml` configuration file, supporting `doom.config.yaml` or `doom.config.yml`. For complex scenarios, such as requiring dynamic configuration or custom `rspress` plugins, `js/ts` configuration files can be used, supporting multiple file formats including `.js/.ts/.mjs/.mts/.cjs/.cts`.

For `js/ts` configuration files, we need to export the configuration, which can be combined with the `defineConfig` function exported from `@alauda/doom/config` to provide type assistance:

```
import { defineConfig } from '@alauda/doom/config'

export default defineConfig({})
```

---

## Basic Configuration

- `lang` : Default documentation language. To accommodate most projects, we support both Chinese and English documents by default. The default language is `en`. If the current documentation project does not require multilingual support, this can be set to `null` or `undefined`.
- `title` : Documentation title, displayed on the browser tab.
- `logo` : Logo at the top left of the documentation, supports image URLs or file paths. Absolute paths refer to files under the `public` directory, relative paths refer to files relative to the current tool directory. By default, the built-in Alauda logo from the `doom` package is used.
- `logoText` : Documentation title, displayed next to the logo at the top left.
- `icon` : Documentation favicon, defaults to the same as `logo`.
- `base` : Base path for the documentation, used when deploying to a non-root path, such as `product-docs`. Defaults to `/`.
- `outDir` : Directory for build outputs, defaults to `dist/{base}/{version}`. If specified, it changes to `dist/{outDir}/{version}`, where `version` is optional. See [Multi-version Build](#) for reference.

## API Documentation Configuration

```
api:
  # CRD definition file paths, relative to the directory of doom.config.*, su
  crds:
    - docs/shared/crds/*.yaml
  # OpenAPI definition file paths, relative to the directory of doom.config.*
  openapis:
    - docs/shared/openapis/*.json
  # When rendering OpenAPI related resource definitions, they are inlined by
  # Reference https://doom.alauda.cn/apis/references/CodeQuality.html#v1alpha
  references:
    v1alpha1.CodeQualityBranch: /apis/references/CodeQualityBranch#v1alpha1.C
  # Optional, API documentation path prefix. If the current business uses gat
  pathPrefix: /apis
```

Refer to [API Documentation](#) for writing documentation.

## Permission Documentation Configuration

```
# The following resource file paths are relative to the directory of doom.con
permission:
  functionresources:
    # `kubectl get functionresources`
    - docs/shared/functionresources/*.yaml
  roletemplates:
    # `kubectl get roletemplates -l auth.cpaas.io/roletemplate.official=true`
    - docs/shared/roletemplates/*.yaml
```

Refer to [Permission Documentation](#) for writing documentation.

## Reference Documentation Configuration



**reference:**

- **repo:** `alauda-public/product-doc-guide` # Optional, repository address of the referenced documentation
- branch:** # [string] Optional, branch of the referenced documentation repository
- publicBase:** # [string] Optional, when using a remote repository, the absolute base path of the repository
- sources:**
  - **name:** `anchor` # Name of the referenced documentation, used for reference
  - path:** `docs/index.mdx#` # Path of the referenced documentation, supports anchors
  - ignoreHeading:** # [boolean] Optional, whether to ignore the heading. If true, the heading will be ignored
  - processors:** # Optional, processors for referenced document content
    - **type:** `ejsTemplate`
    - data:** # ejs template parameters, accessed via `<%= data.xx %>`
  - frontmatterMode:** `merge` # Optional, frontmatter handling mode for referenced document

**frontmatterMode**

- **ignore** : Ignore the frontmatter of the referenced document, keep using the current document's frontmatter.
- **merge** : Merge the frontmatter of the referenced document. If keys conflict, the referenced document's values override the current document's.
- **replace** : Replace the current document's frontmatter with that of the referenced document.
- **remove** : Remove the current document's frontmatter.

Refer to [Reference Documentation](#) for writing documentation.

## Release Notes Configuration

**releaseNotes:****queryTemplates:**

- fixed:** # JQL statements that can include ejs templates
- unfixed:**

release-notes.md

```
<!-- release-notes-for-bugs?template=fixed&project=DevOps -->
```

release-notes.mdx

```
{/* release-notes-for-bugs?template=fixed&project=DevOps */}
```

Taking the above `template=fixed&project=DevOps` as an example, `fixed` is the template name defined in `queryTemplates`. The remaining `query` parameter `project=DevOps` is passed as `ejs` [↗](#) template parameters to the `fixed` template, which after processing is used to form a jira `jql` [↗](#) request to `https://jira.alauda.cn/rest/api/2/search?jql=<jql>`. This API requires authentication, and the environment variables `JIRA_USERNAME` and `JIRA_PASSWORD` must be provided to preview the effect.

## Sidebar Configuration

```
sidebar:
  collapsed: false # Optional, whether the sidebar is collapsed by default. D
```

## Internal Documentation Routes Configuration

```
internalRoutes: # Optional, supports glob matching, relative to the docs dire
  - '*/internal/**'
```

## Only Include Documentation Routes Configuration

```
onlyIncludeRoutes: # Optional, supports glob matching, relative to the docs d
  - '*/internal/**'
internalRoutes:
  - '*/internal/overview.mdx'
```

---

## Syntax Highlighting Plugin Configuration

```
shiki:
  theme: # optional, https://shiki.style/themes
  langs: # optional, https://shiki.style/languages
  transformers: # optional, only available in js/ts config, https://shiki.sty
```

### WARNING

Unconfigured languages will trigger warnings in the command line and fall back to rendering as `plaintext`.

---

## `sites.yaml` Configuration

The `sites.yaml` configuration file is used to configure subsite information associated with the current documentation site. This information is used by [External Site Components](#) and when building single-version documentation.

```
- name: connectors # Globally unique name
base: /devops-connectors # Base path for site access
version: v1.1 # Version used for ExternalSite/ExternalSiteLink redirection

displayName: # Site display name. If not filled or language not matched, de
  en: DevOps Connectors
  zh: DevOps 连接器

# The following properties are used to pull images when building the entire
# Usually required for subsite references, not required for parent site ref
repo: https://github.com/AlaudaDevops/connectors-operator # Site repository
image: devops/connectors-docs # Site build image, used to pull images when
```

---

## Translation Configuration

**translate:**

```
# System prompt, ejs template. Parameters passed in include `sourceLang`, `
# Among them, `sourceLang` and `targetLang` are the strings `[]` and `[]`,
# `userPrompt` is the global user configuration below, which may be empty
# `additionalPrompts` is the `additionalPrompts` configuration in the document
# `terms` and `titleTranslationPrompt` are dynamically generated prompts
# The default system prompt is as follows and can be modified according to
systemPrompt: |
```

You are a professional technical documentation engineer, skilled in writing high-quality technical documentation.

**## Baseline Requirements**

- Sentences should be fluent and conform to the expression habits of the target language.
  - Input format is MDX; output format must also retain the original MDX format.
  - **\*\*CRITICAL\*\*:** Do not translate or modify ANY link content in the document.
    - URLs in markdown links: [text](URL) - keep URL exactly as is
    - Reference-style links: [text][ref] and [ref]: URL - keep both ref and URL
    - Inline URLs: https://example.com - keep completely unchanged
    - Image links: ![alt](src) - keep src unchanged, but alt text can be translated
    - Anchor links: [text](#anchor) - keep #anchor unchanged
    - Any href attributes in HTML tags - keep unchanged
  - Do not translate professional technical terms and proper nouns, including brand names.
  - The title field and description field in frontmatter should be translated.
  - Content within MDX components needs to be translated, whereas MDX component names should not be translated.
  - Do not modify or translate any placeholders in the format of \_\_ANCHOR\_N\_\_ (e.g., \_\_ANCHOR\_1\_\_, \_\_ANCHOR\_2\_\_, etc.).
  - Keep original escape characters like backslash, angle brackets, etc. unchanged.
  - Do not add any escape characters to special characters like [], (), {}, etc.
    - If source has "Architecture [Optional]", keep it as "Architecture [Optional]".
    - If source has "Function (param)", keep it as "Function (param)" (not "Function (param)").
    - Only add escape characters if they were present in the original text.
  - Preserve and do not translate the following comments, nor modify their content:
    - `/* release-notes-for-bugs */`
    - `<!-- release-notes-for-bugs -->`
  - Remove and do not retain the following comments:
    - `/* reference-start */`
    - `/* reference-end */`
    - `<!-- reference-start -->`
    - `<!-- reference-end -->`
  - Ensure the original Markdown format remains intact during translation, such as code blocks, tables, etc.
  - Do not translate the content of the code block.
- ```
<% if (titleTranslationPrompt) { %>
<%- titleTranslationPrompt %>
<% } %>
<% if (terms) { %>
```

```

<%- terms %>
<% } %>

<% if (userPrompt || additionalPrompts) { %>
## Additional Requirements
These are additional requirements for the translation. They should be met also

The text for translation is provided below, within triple quotes:
"""
<% if (userPrompt) { %>
<%- userPrompt %>
<% } %>

<% if (additionalPrompts) { %>
<%- additionalPrompts %>
<% } %>
"""
<% } %>

```

## Editing Documentation in Code Repository

```
editRepoBaseUrl: alauda/doom/tree/main/docs # The https://github.com/ prefix
```

## Documentation Export Configuration

```

export:
- name: Concepts # Optional, globally unique PDF name, defaults to the document name
  scope: '*/concepts' # Required, string or array, document scope, supports

```

## Documentation Lint Configuration

```
lint:  
  cspellOptions: # Optional, cspell configuration options, refer to https://g
```

## Algolia Search Configuration

```
algolia: # Optional, Algolia search configuration, only effective when the CL  
  appId: # Algolia Application ID  
  apiKey: # Algolia API Key  
  indexName: # Algolia index name
```

Please use `public/robots.txt` for Algolia crawler verification.

### INFO

Due to the current architecture limitations of `rspress`, using Algolia search requires implementing via a [custom theme](#). To unify usage of related theme features, we provide the

`@alauda/doom/theme` theme entry. Please add the following theme configuration file to enable:

```
theme/index.ts
```

```
export * from '@alauda/doom/theme'
```

## Sitemap Configuration

```
siteUrl: https://docs.alauda.cn # Optional, site URL used to generate sitemap
```

# Convention

## TOC

[Directory Structure](#)[Metadata](#)[Sorting](#)[Preview](#)

## Directory Structure

The left sidebar is automatically generated based on the file directory structure, where the `index` file in the first-level directory acts as the document's homepage and will display as the first item in the left navigation. Subfolders can use `index.md` or `index.mdx` and define the first-level title to set the grouping title for the left sidebar. Other sub-documents will be automatically merged into the current group, and nested subfolders will follow the same rules.

```
├─ index.md
├─ start.mdx
├─ usage
  └─ index.mdx
     └─ convention.md
```

We also agree that:

1. The `public` directory is used to store static resources such as images, videos, etc.
2. The `public/_remotes` directory is used to store static resources associated with [remote reference documents](#). Please do not directly rely on resources from this directory;



you may add `*/public/_remotes` to `.gitignore` to prevent these from being committed to the code repository.

3. The `shared` directory is for storing common components, reusable documents, etc., and will not automatically generate document data.

---

## Metadata

At the beginning of the document, you can define the document's metadata such as title, description, author, category, etc., through the `frontmatter`.

```
---  
title: Title  
description: Description  
author: Author  
category: Category  
---
```

In the body of the document, when using `.mdx` files, you can access these metadata through `frontmatter` as described in [MDX](#).

---

## Sorting

Other documents, except for `index.md` or `index.mdx`, will be sorted by default according to their file names. You can customize the `weight` value in the `frontmatter` to adjust the order of documents in the left sidebar (the smaller the `weight` value, the higher the priority in sorting).

```
---  
weight: 1  
---
```

**WARNING**

Note: Currently, changes to the left navigation configuration require a service restart to take effect, and it is usually not necessary to pay too much attention during development.

## Preview

Sometimes, we do not need to display special content on the group homepage. In this case, you can use `index.mdx` file and the `Overview` component to display the list of documents in the current group. This will showcase the titles, descriptions, and secondary title information of the grouped list file.

# Usage

```
<Overview />
```

You can refer to [Usage](#) for the effect.

# Markdown

In addition to the standard [gfm](#) syntax, Doom has built-in some extra Markdown extension features.

## TOC

Callouts

Mermaid

Attribute Extensions

## Callouts

Source code annotation components

### NOTE

1. Please use inline code comments according to the actual language, such as `;`, `%`, `#`, `//`, `/** */`, `--`, and `<!-- -->`, etc.
2. If you want to treat them as code comments, please escape them with `[\\!code callout]`.
3. Sometimes, `:::callouts` may display abnormally due to nested indentation; you can use `<div class="doom-callouts">` or the `<Callouts>` component instead.

```
```sh
```

```
Memory overhead per virtual machine ≈ (1.002 × requested memory) \
    + 218 MiB \ # [\\!code callout]
    + 8 MiB × (number of vCPUs) \ # [\\!code callout]
    + 16 MiB × (number of graphics devices) \ # [\\!code callout]
    + (additional memory overhead) # [\\!code callout]
```

```
```
```

```
:::callouts
```

1. Required for the processes that run in the `virt-launcher` pod.
2. Number of virtual CPUs requested by the virtual machine.
3. Number of virtual graphics cards requested by the virtual machine.
4. Additional memory overhead:
  - If your environment includes a Single Root I/O Virtualization (SR-IOV) network device or a Graphics Processing Unit (GPU), allocate 1 GiB additional memory overhead for each device.
  - If Secure Encrypted Virtualization (SEV) is enabled, add 256 MiB.
  - If Trusted Platform Module (TPM) is enabled, add 53 MiB.

```
:::
```

```
Memory overhead per virtual machine ≈ (1.002 × requested memory) \
    + 218 MiB \ ①
    + 8 MiB × (number of vCPUs) \ ②
    + 16 MiB × (number of graphics devices) \ ③
    + (additional memory overhead) ④
```

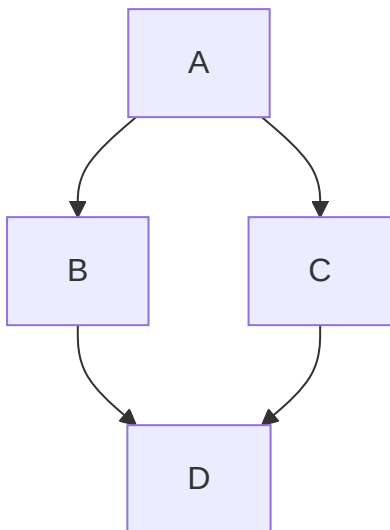
- ① Required for the processes that run in the `virt-launcher` pod.
- ② Number of virtual CPUs requested by the virtual machine.
- ③ Number of virtual graphics cards requested by the virtual machine.
- ④ Additional memory overhead:
  - If your environment includes a Single Root I/O Virtualization (SR-IOV) network device or a Graphics Processing Unit (GPU), allocate 1 GiB additional memory overhead for each device.
  - If Secure Encrypted Virtualization (SEV) is enabled, add 256 MiB.
  - If Trusted Platform Module (TPM) is enabled, add 53 MiB.

For more source code transformation features, please refer to [Shiki Transformers](#) ↗.

## Mermaid [↗](#)

Chart drawing tool

```
```mermaid
graph TD;
  A-->B;
  A-->C;
  B-->D;
  C-->D;
```
```



With [Markdown Preview Mermaid](#) [↗](#), you can preview in real time in VSCode.

## Attribute Extensions

### WARNING

Currently only supported in `.md` files, see [related issue](#) [↗](#)

```
{width="100" height="100"}
```



# MDX

[MDX ↗](#) is an extension syntax of Markdown that allows the use of JSX syntax within Markdown. For usage, refer to [rspress MDX ↗](#).

## TOC

rspress Components

doom Components

Overview

Directive

ExternalSite

ExternalSiteLink

AcpApisOverview and ExternalApisOverview

Term

props

TermsTable

props

JsonViewer

Custom Component Reuse

## rspress Components

Most of the [built-in components ↗](#) provided by the `rspress` theme have been adjusted to global components, which can be used directly in `.mdx` files without importing, including:

- `Badge`

- `Card`
- `LinkCard`
- `PackageManagerTabs`
- `Steps`
- `Tab/Tabs`
- `Toc`

Other less commonly used components can be imported from `rspress/theme`, for example:

preview.mdx

```
import { SourceCode } from '@rspress/core/theme'

<SourceCode href="/" />
```

---

## doom Components

`doom` provides some global components to assist in documentation writing, which can be used directly without import. Currently, these include:

### Overview

Document overview component, used to display the document directory

### Directive

Sometimes, due to nested indentation, the [custom container](#) syntax may fail. You can use the `Directive` component as a substitute.



- The directory structure of multilingual documents (``doc/en``) needs to be ex

```
<Directive type="danger" title="Note">
```

If you are using automated translation tools, you do not need to worry about this issue, as the automated translation tools will automatically generate the target language document directory structure based on ``doc/zh``.

```
</Directive>
```

- The directory structure of multilingual documents ( `doc/en` ) needs to be exactly the same as the documents under the `doc/zh` directory to ensure that the links in multilingual documents are identical except for the language identifier.

### Note

If you are using automated translation tools, you do not need to worry about this issue, as the automated translation tools will automatically generate the target language document directory structure based on `doc/zh`.

## ExternalSite

Component for referencing external sites

```
<ExternalSite name="connectors" />
```

### Note

Because DevOps Connectors releases on a different cadence from Alauda Container Platform, the DevOps Connectors documentation is now available as a separate documentation set at [DevOps Connectors ↗](#).

## ExternalSiteLink

Component for referencing external site links

```
<ExternalSiteLink name="connectors" href="link.mdx#hash" children="Content" />
```

## Content ↗

### TIP

In mdx, `<ExternalSiteLink name="connectors" href="link" children="Content" />` has a different meaning from the following:

```
<ExternalSiteLink name="connectors" href="link">
  Content { /* This will be rendered inside a `p` element */ }
</ExternalSiteLink>
```

If you do not want the text to be rendered inside a `p` element, you can pass it via the `children` prop as shown in the example above.

## AcpApisOverview and ExternalApisOverview

Components for referencing external site API overviews

```
<AcpApisOverview />
{ /* same as following */ }
<ExternalApisOverview name="acp" />

<ExternalApisOverview name="connectors" />
```

### Note

For the introduction to the usage methods of ACP APIs, please refer to [ACP APIs Guide ↗](#).

### Note

For the introduction to the usage methods of DevOps Connectors APIs, please refer to [DevOps Connectors APIs Guide ↗](#).

## Term

Term component, plain text, dynamically mounted and injected

```
<Term name="company" textCase="capitalize" />
<Term name="product" textCase="lower" />
<Term name="productShort" textCase="upper" />
```

Alauda alauda container platform ACP

## props

- `name` : Built-in term name, refer to [dynamic mounting configuration file](#)
- `textCase` : Text case transformation, optional values are `lower` , `upper` , `capitalize`

## TermsTable

Built-in term list display component

```
<TermsTable />
```

| Name         | Chinese     | Chinese<br>Bad<br>Cases | English                         | English<br>Bad<br>Cases | Description |
|--------------|-------------|-------------------------|---------------------------------|-------------------------|-------------|
| company      | 公司          | -                       | Alauda                          | -                       | 公司          |
| product      | 容器平台<br>ACP | -                       | Alauda<br>Container<br>Platform | -                       | 容器平台        |
| productShort | ACP         | -                       | ACP                             | -                       | 容器平台        |

## props

- `terms` : `NormalizedTermItem[]` , optional, custom term list to facilitate reuse when rendering custom terms in internal documentation

## JsonViewer

```
<JsonViewer value={{ key: 'value' }} />
```

```
yaml    json
```

```
key: value
```

## Custom Component Reuse

According to the [convention](#), we can extract reusable content into the `shared` directory and then import it where needed, for example:

```
import CommonContent from './shared/CommonContent.mdx'

<CommonContent />
```

If you need to use more [runtime](#) related APIs, you can implement components in `.jsx/.tsx` and then import and use them in `.mdx` files.

```
// shared/CommonContent.tsx
export const CommonContent = () => {
  const { page } = usePageData()
  return <div>{page.title}</div>
}

// showcase/content.mdx
import { CommonContent } from './shared/CommonContent'
;<CommonContent />
```

### WARNING

Note: Currently, components exported from `.mdx` do not support passing `props`. Refer to [this issue](#). Therefore, for scenarios requiring `props` passing, please develop using `.jsx/.tsx`

components.

# Internationalization

Most of the internal documentation of `alauda` is bilingual in Chinese and English. Therefore, we by default support using `en / zh` subfolders to store documents in different languages. It is recommended to also store static resources under `en / zh` subfolders in the `public` directory, which facilitates the management of both documentation content and static resources.

## TOC

`i18n.json``.ts/.tsx``.mdx`

## `i18n.json`

For reusable components, if you need to support both Chinese and English within the same component, you first need to create an `i18n.json` file under the `docs` directory. Then, in the component, use `useI18n` to get the text for the current language, for example:

`docs/i18n.json`

```
{
  "title": {
    "zh": "标题",
    "en": "Title"
  },
  "description": {
    "zh": "描述",
    "en": "description"
  }
}
```

---

## .ts/.tsx

```
import { useI18n } from '@rspress/core/runtime'

export const CommonContent = () => {
  const t = useI18n()
  return <h1>{t('title')}</h1>
}
```

---

## .mdx

```
import { useI18n } from '@rspress/core/runtime'

# {useI18n()}('title')}

{useI18n()}('description')}
```

# API Documentation

Based on actual business needs, we generally categorize APIs into two types: Advanced APIs and CRDs (Custom Resource Definitions). Therefore, the directory structure is typically organized as follows:

```
|— apis
|   |— advanced-apis # Advanced APIs
|   |— crds # CRDs
|   |— references # Common References
```

## TOC

Advanced APIs

props

CRD

props

Common References

props

Specifying OpenAPI Path

## Advanced APIs

advanced-apis/codeQualityTaskSummary.mdx



```
# CodeQualityTaskSummary
```

```
<OpenAPIPath path="/plugins/v1alpha1/template/codeQuality/task/{task-id}/summ
```

Refer to [CodeQualityTaskSummary](#).

## props

- `path` : The path under OpenAPI schema `paths`
- `pathPrefix` : Can be used to override the `api.pathPrefix` in global configuration
- `openapiPath` : Refer to [Specifying OpenAPI Path](#)

## CRD

```
crds/ArtifactCleanupRun.mdx
```

```
# ArtifactCleanupRun
```

```
<K8sCrd name="artifactcleanupruns.artifacts.katanomi.dev" />
```

Refer to [ArtifactCleanupRun](#).

## props

- `name` : CRD `metadata.name`
- `crdPath` : Similar to [Specifying OpenAPI Path](#), used to specify a particular CRD file

## Common References

```
references/CodeQuality.mdx
```

```
# CodeQuality
```

```
<OpenAPIRef schema="v1alpha1.CodeQuality" />
```

Refer to [CodeQuality](#).

## props

- `schema` : The name under OpenAPI schema `definitions` (v2) or `components/schemas` (v3)
- `openapiPath` : Refer to [Specifying OpenAPI Path](#)

---

## Specifying OpenAPI Path

For the `OpenAPIPath` and `OpenAPIRef` components, the default behavior is to search for matches across all OpenAPI definition files. If you need to specify a particular OpenAPI file, you can use the `openapiPath` property:

```
<OpenAPIPath  
  path="/plugins/v1alpha1/template/codeQuality/task/{task-id}/summary"  
  openapiPath="shared/openapis/katanomi.json"  
>
```

# Permission Description Document

```
<K8sPermissionTable functions={['devops-testplans', 'devops-testmodules']} />
```

## TOC

props

Example

### props

- functions** : `string[]` - Required. An array of `FunctionResource` resource names to be displayed.

## Example

| Function                      | Action | Platform Administrator | Platform auditors | Project Manager | Namespace Administrator |
|-------------------------------|--------|------------------------|-------------------|-----------------|-------------------------|
| testplans<br>devops-testplans | View   | ✓                      | ✓                 | ✓               | ✓                       |
|                               | Create | ✓                      | ✗                 | ✓               | ✓                       |
|                               | Update | ✓                      | ✗                 | ✓               | ✓                       |
|                               | Delete | ✓                      | ✗                 | ✓               | ✓                       |

| Function                               | Action | Platform Administrator | Platform auditors | Project Manager | Namespace Administrator |
|----------------------------------------|--------|------------------------|-------------------|-----------------|-------------------------|
| testmodules<br>devops -<br>testmodules | View   | ✓                      | ✓                 | ✓               | ✓                       |
|                                        | Create | ✓                      | ✗                 | ✓               | ✓                       |
|                                        | Update | ✓                      | ✗                 | ✓               | ✓                       |
|                                        | Delete | ✓                      | ✗                 | ✓               | ✓                       |

# Referencing Documents

In Markdown files:

```
<!-- reference-start#name -->
```

```
<!-- reference-end -->
```

In MDX files:

```
{/* reference-start#name */}
```

```
{/* reference-end */}
```

The `name` above refers to the name of the referenced document. For more information, please refer to [Document Reference Configuration](#). If the referenced document content uses static resources from a remote repository, the related static resources will be automatically stored locally in the `<root>/public/_remotes/<name>` directory.

Here is an example using `<!-- reference-start#ref -->`:

---

## TOC

Document Reference Configuration

`frontmatterMode`

---

## Document Reference Configuration

**reference:**

- **repo:** `alauda-public/product-doc-guide` # Optional, repository address for
- branch:** # [string] Optional, branch of the referenced document repository
- publicBase:** # [string] Optional, the directory where static resources for
- sources:**
  - **name:** `anchor` # Name of the referenced document, used to reference with
  - path:** `docs/index.mdx#introduction` # Path to the referenced document,
  - ignoreHeading:** # [boolean] Optional, whether to ignore headings. If t
  - processors:** # Optional, processors for handling the content of the re
    - **type:** `ejsTemplate`
    - data:** # EJS template parameters, accessed via `<%= data.xx %>`.
  - frontmatterMode:** `merge` # Optional, mode for handling the frontmatter

**frontmatterMode**

- `ignore` : Ignores the frontmatter of the referenced document and retains the frontmatter of the current document.
- `merge` : Merges the frontmatter of the referenced document. If there are the same keys, the values from the referenced document will overwrite those in the current document.
- `replace` : Replaces the frontmatter of the current document with that of the referenced document.
- `remove` : Removes the frontmatter of the current document.

For writing documentation, refer to [Document Reference](#).

# Deployment

## TOC

[Build and Preview](#)[Multi-version Build](#)[Merged Directory Structure](#)[Dynamic Mount Configuration File](#)

## Build and Preview

Before deployment, we need to build the project for production and preview it locally to ensure the project runs correctly:

```
doom build # Build static assets
doom serve # Preview the build assets in production mode
```

## Multi-version Build

By default, `doom build` outputs the build artifacts to the `dist` directory. If you need to build multiple versions of the documentation, you can specify the version number with the `-v` parameter, for example:

```
# Usually determined by branch name, e.g., release-4.0 corresponds to version
doom build -v 4.0 # Build version 4.0, output to dist/4.0, documentation acce
doom build -v master # Build master version, output to dist/master, documenta
doom build -v {other} # Build other versions, output to dist/{other}, documen

# unversioned and unversioned-x.y are special version numbers used to build d
doom build -v unversioned # Build documentation without version prefix, output
doom build -v unversioned-4.0 # Build documentation without version prefix bu
```

## Merged Directory Structure

```
|— console-platform
|   |— 4.0
|   |— 4.1
|   |— index.html
|   |— overrides.yaml
|   └─ versions.yaml
|— console-devops-docs
|   |— 4.0
|   |— 4.1
|   |— index.html
|   |— overrides.yaml
|   └─ versions.yaml
|— console-tekton-docs
|   |— 1.0
|   |— 1.1
|   |— index.html
|   |— overrides.yaml
|   └─ versions.yaml
```

index.html



```
<!DOCTYPE html>
<html>
  <head>
    <title>Redirecting...</title>
    <meta http-equiv="refresh" content="0; url=/console-docs/4.1" />
  </head>
  <body>
    <p>Redirecting to <a href="/console-docs/4.1">/console-docs/4.1</a></p>
  </body>
</html>
```

## Dynamic Mount Configuration File

overrides.yaml

```
# Documentation information, each document can mount to override default conf
title:
  en: Doom - Alauda
  zh: Doom - 道达
logoText:
  en: Doom - Alauda
  zh: Doom - 道达
```

versions.yaml

```
- '4.1'
- '4.0'
```